

0 AND 1: The Thaumaturgical Correlation

0: THE ARCHETYPE OF ABSENCE

ALGEBRA: 0 is the additive identity ($0+x=x$) and the annihilator ($0*x=0$).

0 is the element that CHANGES NOTHING and DESTROYS everything.

ANALYSIS: 0 is the singularity of $\log$ ($\log(0)=-\infty$) and the pole of $1/x$ ($1/0=\infty$).

0 is the gateway to the negative infinite and the generator of the infinite.

GEOMETRY: 0 is the origin of $\mathbb{R}$, the boundary between positive and negative.

0 is the EVENT HORIZON of the real line.

SET THEORY: $|\{\}| = 0$. The empty set has no elements.

0 is the cardinality of NOTHINGNESS. The foundation of ZFC.

PHYSICS: The vacuum $|0\rangle$ has no particles but HAS energy (zero-point).

0 (no particles) $\neq$ 0 (no energy). The vacuum is not empty.

COSMOLOGY: At $t=0$: $a=0$, $\rho=\infty$, $T=\infty$. The Big Bang singularity.

0 is the moment of CREATION and DESTRUCTION.

1: THE ARCHETYPE OF UNITY

ALGEBRA: 1 is the multiplicative identity ($1 \cdot x = x$).

1 is the element that CHANGES NOTHING (multiplicatively).

1 is the ATOM of the number system (generator of $\mathbb{Z}$).

1 is IMMUTABLE: $1^n = 1$ for all n .

ANALYSIS: $e^0 = 1$. The exponential sends $0 \rightarrow 1$.

1 is the IMAGE of the void under $\exp$.

$[0,1]$ is the fundamental domain of probability. 1 = CERTAINTY.

GEOMETRY: 1 is the identity of $\mathbb{R}^*$. The FIXED POINT of $x \rightarrow 1/x$.

SET THEORY: $|\{x\}| = 1$ for any x .

1 is the cardinality of EXISTENCE (the singleton has content).

PHYSICS: $U(1)$ has dimension 1. The Abelian symmetry.

The fine structure constant $\alpha \sim 1/137$.

COSMOLOGY: de Sitter space: $a(t) = \exp(Ht)$, $a(0) = 1$.

1 is the INITIAL SIZE of the universe (no singularity).

THE COLLISION: 0 AND 1

$0 + 1 = 1$: PRESERVATION (the void cannot destroy what exists)

$0 * 1 = 0$: DESTRUCTION (the void destroys multiplicatively)

$1 / 0 = \text{inf}$: GENERATION (the void creates the infinite)

$0 / 0 = ?$: CHOICE (the removable singularity: outcome is open)

$0^0 = 1$: CREATION (the void raised to unity creates)

$0^x = 0$: ANNIHILATION (the void raised to existence destroys)

$x^0 = 1$: IDENTITY (existence raised to void preserves)

THE THREE ZEROS:

$\zeta(0) = -1/2$ $\sin(0)*\zeta(1) = 0*\text{inf} = -1/2$ (removable)

$\zeta(-1) = -1/12$ $1+2+3+\dots$ diverges, regularized to $-1/12$

$\zeta(-2n) = 0$ sin factor gives trivial zeros

THE THREE ONES:

$e^0 = 1$ the exponential of the void

$0! = 1$ the factorial of the void

$\sin(0)/0 = 1$ the sinc function at the origin

BRANCH 1: VON NEUMANN ALGEBRAS

[von Neumann 1932; Connes 1976; Haag 1992]

Factors classify von Neumann algebras by their trace structure:

Type I_n/I_∞: $B(H)$. Center = $\{\lambda \cdot I\}$. 0/0 is TRIVIAL.

Type II₁: Tracial state $\tau(I)=1$. 0/0 is RESOLVED by the trace.

Type II_∞: Unbounded trace. $\tau(0) \cdot \tau(I) = 0 \cdot \infty$.

This is EXACTLY $\sin(0) \cdot \zeta(1) = 0 \cdot \infty$.

Type III: No trace exists. 0/0 is IRREDUCIBLE.

QFT uses Type III factors [Haag 1992].

The vacuum expectation value: $\langle 0|I|0\rangle = 1$.

$|0\rangle$ is '0' (no particles). I is '1' (identity).

$0^0 = 1$: the vacuum gives unity.

This is the PHYSICAL realization of $0^0 = 1$.

BRANCH 2: CONNES' NONCOMMUTATIVE GEOMETRY

[Connes 1994; Connes-Marcolli 2008; Atiyah-Singer 1968]

Spectral triple (A, H, D) : algebra, Hilbert space, Dirac operator.

The spectral action: $S = \text{Tr}(f(D/\Lambda))$.

At $D=0$: $\text{Tr}(f(0)) = f(0) \cdot \dim(H) = \infty \cdot f(0)$. This is $0 \cdot \infty$ or $\infty \cdot \text{finite}$.

Connes resolves via ZETA REGULARIZATION:

$$\det(D) = \exp(-\zeta'_D(0))$$

$$\zeta_D(s) = \sum \lambda_n^{-s} \text{ (spectral zeta function)}$$

The spectral zeta GENERALIZES $\zeta(s)$:

$$\text{For } S^1: \zeta_D(s) = \zeta(s) \cdot (2\pi)^{-s}$$

$$\zeta_D(0) = -\dim(M)/2$$

$$\zeta(0) = -1/2 \text{ for } S^1. \quad \zeta(0) = -3/2 \text{ for } S^3.$$

The Standard Model [Connes-Marcolli 2008]:

$$A = C^\infty(M) \otimes C^\infty(\mathbb{R} \times C_8)$$

The $0/0$ is the product of continuous and finite.

BRANCH 3: AdS/CFT AND HOLOGRAPHY

[Maldacena 1998; Witten 1998; Bekenstein 1973; Hawking 1975]

AdS/CFT: IIB string on $AdS_5 \times S^5 \iff N=4$ SYM on $R^{1,3}$.

The holographic principle: information in a volume is on its boundary.

$S_{BH} = A/(4G)$ [Bekenstein 1973].

The 0/0 of holography:

Bulk metric $\rightarrow 0$ at boundary (degenerate).

Boundary theory is FINITE (the CFT).

0 (degenerate metric) $\rightarrow$ finite (CFT).

Thermal 0/0:

$T_H = 1/(8\pi M)$. At $M=0$: $T=\infty$ (annihilation).

$S_{BH} = 4\pi M^2$. At $M=0$: $S=0$.

But microstates = 1 (the vacuum).

$0^0 = 1$: zero mass, one microstate.

BRANCH 4: L-FUNCTIONS AND THE LANGLANDS PROGRAM

[Riemann 1859; Euler 1737; Grothendieck 1960s; Bost 1992]

$$\zeta(s) = \sum n^{-s} = \prod_p (1 - p^{-s})^{-1}$$

The Euler product connects analysis (sum) to arithmetic (primes).

At $s=0$: each factor = $1/(1-1) = 1/0 = \text{inf}$. Product diverges.

But $\zeta(0) = -1/2$. The $0/0$ is regularized.

Atiyah-Singer [1968]: $\text{ind}(D) = \text{integral of } \hat{A} \text{ genus}$.

The Lefschetz number: $L(f) = \sum (-1)^i \text{Tr}(f^*)$.

This is a $0/0$: alternating sum of traces.

The removable value is the Euler characteristic.

Langlands program: automorphic $\leftrightarrow$ Galois representations.

$L(0, \pi) = L(0, \rho)$. Both diverge. Removable value = invariant.

Grothendieck motives [1960s]:

$h(\text{point}) = 1$. $h(\text{empty}) = 0$.

$L(s, \text{point}) = \zeta(s)$. $L(s, \text{empty}) = 1$.

$\zeta(0) = -1/2$ is the $0/0$ of the point's L-function.

BRANCH 5: QUANTUM INFORMATION

[von Neumann 1932; Bell 1964]

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$.

If $\alpha=\beta=0$: null vector. NORMALIZATION FORBIDS THE 0/0.

Quantum mechanics prevents the 0/0 of absence meeting absence.

Bell state partial trace: off-diagonal terms give $0 \cdot 1 = 0$.

But ENTANGLEMENT is in the off-diagonal terms.

The 0/0: entanglement EXISTS in the annihilation.

THE COMPLETE CORRELATION

0 correlates to:	1 correlates to:
absence	unity
destruction	generation
singularity	immutability
origin / vacuum	certainty / existence
Big Bang	de Sitter
Type III (no trace)	Type II_1 (tracial)
boundary (degenerate)	bulk (finite)
trivial motive ($h=0$)	unit motive ($h=1$)

THE 0/0 FRAMEWORK:

The study of what happens when ABSENCE meets ABSENCE.

The removable value depends on the context.

The context is the mathematical or physical structure.

The removable value is the MEANING.